CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of BSc (Biological Science) Examination, March 2018

BE112 Biotechnology and Human Welfare

Date: 29/03/2018 Day: Thursday Time: 10.00 a.m. To 1.00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	Select the odd one out related to 'detergent application' (i) Keratinases (ii) Lipases (iii) Proteases (iv) Amylases	
2.	*Bacillus thuringiensis* destroys (i) Rodents (ii) Insects (iii) Fungi (iv) None of these	
3.	The DNA fingerprint pattern of a child matches to (i) Father's DNA pattern only (ii) Mother's DNA pattern only (iii) 50% of father's DNA and rest of mother's DNA (iv) None of the above	
4.	PCR technique involves application of the enzyme Taq Polymerase. □True □False	
5.	Terminator technology produces fertile seeds. □True □False	

6.	A ‘biotechnological process’ involves __________ as its first step. (i) Mass culture (ii) Cell response optimization (iii) Identification of organism or cell type (iv) None of the above	
7.	Production of monoclonal antibodies involves cell culture and cell fusion □True □False	
8.	Application of biotechnology involving improvement in agriculture sector is ___ (i) Red biotechnology (ii) Green biotechnology (iii) Blue biotechnology (iv) White biotechnology	
9.	Humic acid is present in (i) Plants (ii) soil humus (iii) fungi (iv) none of these	
10.	Bioremediation is the application of _______ (i) medical biotechnology (ii) industrial biotechnology (iii) forensic (iv) none of these	
11.	PSM solubilizes which mineral ion? (i) P (ii) K (iii) Fe (iv) none of these	
12.	DST stands for Department of Science and Technology □True □False	
13.	One of the given sequences is the correct order for DNA finger printing procedure (i) DNA isolation → electrophoresis → restriction digestion → probe hybridization → southern blotting →autoradiograph (ii) DNA isolation → restriction digestion → electrophoresis → southern blotting → probe hybridization →autoradiograph (iii) restriction digestion → DNA isolation → electrophoresis → southern blotting → probe hybridization →autoradiograph (iv) DNA isolation → electrophoresis → restriction digestion → southern blotting → probe hybridization →autoradiograph	
14.	Which of the given genes are involved in N fixation? (i) *Nif* genes (ii) ADA genes (iii) SCID genes (iv) None of these	

15.	DNA finger printing was developed by ______. (i) Alec Jeffreys (ii) Rosalind Franklin (iii) Har Gobind Khorana (iv) James Watson	
16.	PHB is a microbial product produced during physiological stress. □True □False	
17.	Arbuscule formation takes place in endomycorrhizae. □True □False	
18.	______ is responsible for increasing surface area for absorption of soil water & nutrients in plant roots. i) mycorrhizae ii) lichens iii) nematodes iv) algae	
19.	Insulin is a (i) carbohydrate (ii) lipid (iii) hormone (iv) none of these	
20.	Biosensor is composed of biological sensing material and an electrode known as a ______ (i) transducer (ii) chip (iii) sensor (iv) none of these	

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Date: 29/03/2018 Day: Thursday Time: 10.00 a.m. To 1.00 p.m Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed		**20**
Q - II A	Answer **any four** of the following:	**08**
(i)	What is a livestock? Which qualitative characters of livestock are improved through biotechnological tools?	
(ii)	Write a brief note on 'Biolistic method' of gene transformation.	
(iii)	How is red biotechnology different from blue biotechnology?	
(iv)	Why is nitrogen fixation considered as energy demanding process?	
(v)	How do plants support the growth of beneficial microbes in rhizosphere (root region)?	
Q - II B	Answer **any three** of the following:	**12**
(i)	Describe the applications of medical biotechnology in medical industry.	
(ii)	Explain in detail about the steps involved in any biotechnological process.	
(iii)	Discuss about 'protein engineering' in detail.	
(iv)	Discuss in detail the steps involved in monoclonal antibody production.	
SECTION - II		
Q-III Answer the following questions as directed		**30**
Q - III A	Answer **any five** of the following:	**10**
(i)	Write a short note on DNA vaccine.	
(ii)	What is a biodegradable polymer? Write two examples.	

(iii)	How can biotechnological tools be applied in agriculture field?	
(iv)	Write a short note on ‘solution of environmental problems through Biotechnological tools’.	
(v)	Explain the process of mycorrhiza formation in plant roots.	
(vi)	Briefly describe the way a live recombinant vaccine works.	
Q - III B	Answer **any five** of the following:	**20**
(i)	Describe the importance of forensic science in determining paternity and maternity.	
(ii)	Give diagrammatic representation of the biological nitrogen fixation (BNF). Also write down the importance of microbe in BNF process.	
(iii)	Write the steps of nodule formation in a legume root. Give suitable diagrams.	
(iv)	Define DNA fingerprinting. Give various applications of DNA fingerprinting.	
(v)	Give a detailed note on the applications of Gene Therapy.	
(vi)	Explain the role of Biotechnology in Dairy industry.	